

Question 1: What is a CSS selector? Provide examples of element, class, and ID selectors.

- A **CSS selector** is used to select the HTML element(s) you want to style.
→ It tells the browser **which elements** the CSS rules should apply to.

- **Examples:**

- **Element Selector:**

- Selects all elements of a specific type.
- Example:
`p { color: blue; }`

→ Changes color of all <p> elements to blue.

- **Class Selector:**

- Used to select elements with a specific class attribute.
- Example:
`.highlight { background-color: yellow; }`

→ Styles all elements with class="highlight".

- **ID Selector:**

- Used to select a single element with a specific id attribute.
- Example:
`#main { font-size: 18px; }`

→ Styles the element with id="main".

Question 2: Explain the concept of CSS specificity. How do conflicts between multiple styles get resolved?

- **Specificity** determines which CSS rule is applied when multiple rules target the same element.

→ The **more specific** selector has **higher priority**.

- **Order of Specificity (High → Low):**

1. **Inline styles** (inside HTML tag) → Highest priority
2. **ID selectors** (#id)
3. **Class, attribute, and pseudo-class selectors** (.class, [type], :hover)

4. **Element selectors** (p, div, h1) → Lowest priority
- When multiple rules apply, **the one with higher specificity wins**.
 - If specificity is the same, the **last rule written in CSS** is applied.

Question 3: What is the difference between internal, external, and inline CSS? Discuss the advantages and disadvantages of each approach.

- Difference between Internal, External, and Inline CSS

1. Inline CSS

- Written inside an HTML tag using the style attribute.
 - Example: `<h1 style="color:red;">Hello</h1>`
- **Advantages:**
 - Quick and easy for small changes.
 - Applies only to a single element.
- **Disadvantages:**
 - Not reusable.
 - Makes HTML code lengthy and hard to maintain.

2. Internal CSS

- Written inside the `<style>` tag within the `<head>` section of an HTML page.
- **Advantages:**
 - Useful for styling a single page.
 - No need for an external file.
- **Disadvantages:**
 - Cannot be reused across pages.
 - Increases page size and load time.

3. External CSS

- Written in a separate .css file and linked using `<link rel="stylesheet" href="style.css">`.
- **Advantages:**
 - Reusable for multiple pages.
 - Keeps HTML clean and easy to maintain.
- **Disadvantages:**

- Needs an extra file to load.
- Styles won't appear if the CSS file fails to load.